TIME USAGE MODEL SPECIFICATION

PRC-OPS-COR-SPC-0001

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1 PURPOSE

The purpose of this document is to define standard performance measures to be applied consistently across all CSI Operations. The performance measures exist to be a basic yet effective method of tracking production losses to allow people to understand and reduce losses and to improve the accuracy of planning activities. These measures will give a clear understanding of whether the Asset Operation is performing to set business targets and also achieving industry benchmarks.

2 SCOPE

The scope of this document is to ensure consistent application of a common time usage model and metrics to improve asset performance in a process driven framework.

The framework of this Time Usage Model (TUM) gives ability to drill down and understand the influences and causes of lost time. Specifically, this document will:

- Define the TUM and its components to be used consistently across CSI Operations.
- Define the accountability of each area of the TUM.
- Describe how to collect and track time usage data via CSI Systems.
- Define how to calculate standard performance measures relating to Time Usage.
- Supply a framework for analysing time usage data and identifying areas for production improvement via cause-and-reason delay accounting specifications.

3 RESPONSIBILITIES

- **Chief Operating Officer:** Responsible for developing, documenting and managing change of the Time Usage Model and Metrics that are consistently applied across the CSI Operations.
- **General Managers & Operations Managers:** Responsible for ensuring the Time Usage Model and Metrics are consistently applied across the CSI Operations, setting suitable targets for each site in line with the CSI Business requirements and deviations from the targets are investigated with corrective actions implemented where required.
- **Site Managers:** Responsible for correctly utilising the Time Usage Model and Metrics to monitor and report on individual site performance, determine non-compliances and implement corrective actions as necessary in conjunction with the Operation Manager.

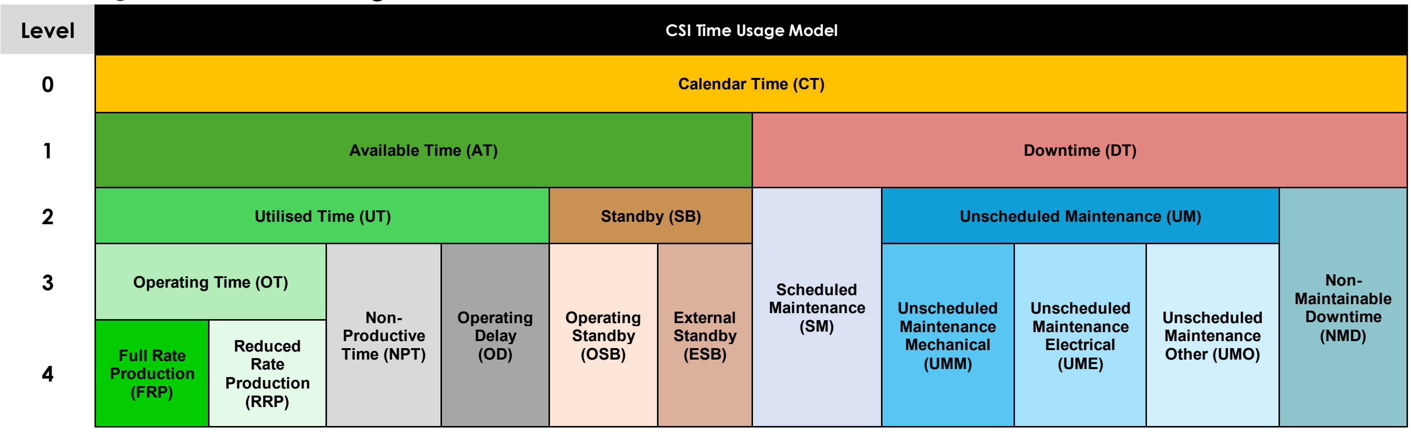
4 TIME USAGE MODEL

The Time Usage Model (TUM) presented in this document has been designed for application across all CSI Mining Services operations and activities. This includes the following areas:

- Crushing & Fixed Plant Operations
- Haulage
- Marine

Figure 1 below shows a visual representation of the CSI TUM, which depicts the time breakdown classifications defined for time and delay accounting purposes.

Figure 1. CSI Time Usage Model



4.1 TUM Definitions

Table 1 below outlines the TUM definitions and associated calculations for each classification. The relevant colour coding should also be used across any visualization or reporting of TUM data when used or distributed by the operation.

Table 1. CSI Time Usage Model Definitions

TUM Level	TUM Classification	TUM Code	Definition	Calculation
Level 0	Calendar Time	CT	Total calendar hours across the period (24 hours/day, 8760 hour/yr).	*Calendar Time (CT) = 24 x #days in period*
Level 1	Available Time	AT	Time the equipment is available to operate for production purposes.	*Available Time (AT) = Calendar Time (CT) – Downtime (DT)*
Level 1	Downtime	DT	Time that the equipment is not capable of being utilised for production.	*Downtime (DT) = Scheduled Maintenance (SM) + Unscheduled Maintenance (BM) + Non-Maintainable Downtime (NMD)*
Level 2	Utilised Time	UT	Time that the equipment is being utilised for production purposes, inclusive of operational delays.	*Utilised Time (UT) = Available Time (AT) – Standby (SB)*
Level 2	Standby	SB	Time that the equipment is available but not utilised.	*Standby (SB) = Operating Standby (OSB) + External Standby (ESB)*
Level 2	Scheduled Maintenance	SM	Time that equipment is scheduled to be unavailable for production purposes due to any scheduled maintenance activities, including inspection, adjustments or equipment maintenance. Planned activities are scheduled in advance of the asset being unavailable.	*Scheduled Maintenance (SM) = SUM (SM events in period)*
Level 2	Unscheduled Maintenance	UM	Time the equipment is out of service / unavailable due to any type of maintenance required due to a breakdown event.	*Unscheduled Maintenance (UM) = Unscheduled Mechanical (UMM) + Unscheduled Electrical (UME) + Unscheduled Other (UMO)*
Level 2	Non-Maintainable Downtime	NMD	Time the equipment is no longer functional, and requires NON-STANDARD maintenance or repairs to return it to a functioning state.	*Non-Maintainable Downtime (NMD) = SUM (NMD events in period)*
Level 3	Operating Time	OT	Time the equipment is available and is being utilised for production purposes, irrespective of its operating rate.	*Operating Time (OT) = Utilised Time (UT) – Non-Productive Time (NPT) – Operating Delay (OD)*
Level 3	Non-Productive Time	NPT	Time the equipment is available and is being utilised but is not contributing to production.	*Non-Productive Time (NPT) = SUM (NPT events in period)*
Level 3	Operating Delay	OD	Time the equipment is available and is being utilised for production purposes but is temporarily stopped or prevented from operating.	*Operating Delay (OD) = SUM (OD events in period)*

TUM Level	TUM Classification	TUM Code	Definition	Calculation
Level 3	Operating Standby	OSB	Time the equipment is available but is not utilised for production purposes due to operational influences. There is no immediate intent to operate due to a management decision or reasons within management control.	***Operating Standby (OSB) = SUM (OSB events in period)***
Level 3	External Standby	ESB	Time the equipment is available but is not utilised for production purposes due to external influences. It cannot be operated for reasons that are outside of the immediate influence of operational management & / or control.	***External Standby (ESB) = SUM (ESB events in period)***
Level 3	Unscheduled Maintenance Mechanical	UMM	Time the equipment is out of service / unavailable due to ***Mechanical*** maintenance required due to a breakdown event.	***Unscheduled Maintenance Mechanical (UMM) = SUM (UMM events in period)***
Level 3	Unscheduled Maintenance Electrical	UME	Time the equipment is out of service / unavailable due to ***Electrical*** maintenance required due to a breakdown event.	***Unscheduled Maintenance Electrical (UME) = SUM (UME events in period)***
Level 3	Unscheduled Maintenance Other	UMO	Time the equipment is out of service / unavailable due to ***Other*** maintenance required due to a breakdown event.	***Unscheduled Maintenance Other (UMO) = SUM (UMO events in period)***
Level 4	Full Rate Production	FRP	Time the equipment is being utilised at or above its defined full production rate.	***Full Rate Production (FRP) = SUM (FRP events in period)***
Level 4	Reduced Rate Production	RRP	Time the equipment is being utilised, but the production rate is below its defined full production rate.	***Reduced Rate Production (RRP) = SUM (RRP events in period)***

5 PERFORMANCE METRICS

5.1 Metrics Definitions

Table 2 below outlines the standard performance metric definitions and associated calculations that are to be used as part of performance monitoring and reporting of CSI operational sites.

Metrics are grouped into levels (i.e. 1-3) according to detail and the implementation will be operations specific and dependent on the defined technical performance reporting requirements.

Table 2. CSI TUM Performance Metrics Definitions

Metric Level	Metric Name	Unit	Definition	Calculation
1	**Availability**	%	The percentage of calendar time the asset is available to perform its intended function, over a defined calendar period.	***Availability (%) = Available Time (AT)/Calendar Time (CT)***
1	**Utilisation**	%	The time the asset is being utilised as a percentage of time over a defined calendar period.	***Utilisation (%) = Utilised Time (UT) / Calendar Time (CT)***
1	**Overall Utilisation**	%	The time the asset is actually operating as a percentage of time over a defined calendar period.	***Overall Utilisation (%) = Operating Time (OT) / Calendar Time (CT)***
1	**Mean Time Between Failure (MTBF)**	hrs	The average time between breakdown events (Utilised Time).	***MTBF (hrs) = Utilised Time (UT) / Count of Unscheduled Maintenance Events (UM)***
1	**Mean Time to Repair (MTTR)**	hrs	The average time taken to repair the total number of all downtime events (Total Downtime).	***MTTR (hrs) = Downtime (DT) / Count of Downtime Events (SM + UM + NMD)***
2	**Mean Time Between Stoppages (MTBS)**	hrs	The average time between operating delay (OD) and non-productive time (NPT) events (Operating Time).	***MTBS (hrs) = Operating Time (OT) / Count of Operating Delay & Non-Productive Time Events (OD + NPT)***

Metric Level	Metric Name	Unit	Definition	Calculation
2	**Utilisation of Available Time**	%	The time the asset is utilised as a percentage of the available time.	***Utilisation of Available Time (%) = Utilised Time (UT) / Available Time (AT)***
2	**Operating Efficiency**	%	The time the asset is actually operating as a percentage of the time that it is being utilised.	***Operating Efficiency (%) = Operating Time (OT) / Utilised Time (UT)***
2	**Standby Rate**	%	The percentage of Standby time over a defined calendar period.	***Standby Rate (%) = Standby Time (SB) / Calendar Time (CT)***
2	**Standby Ratio**	%	The percentage of Standby time out of the total available time.	***Standby Ratio (%) = Standby Time (SB) / Available Time (AT)***
2	**Non-Productive Time Ratio**	%	The percentage of Non-Productive time out of the utilised time.	***Non-Productive Time Ratio (%) = Non-Productive Time (NPT) / Utilised Time (UT)***
2	**Operating Delay Ratio**	%	The percentage of Operating Delay time out of the utilised time.	***Operating Delay Ratio (%) = Operating Delay Time (OD) / Utilised Time (UT)***
2	**Production Efficiency**	%	The percentage of Full Rate Production time out of the total operating time.	***Production Efficiency (%) = Full Rate Production Time (FRP) / Operating Time (OT)***
3	**Scheduled Maintenance Rate**	%	The percentage of Scheduled Maintenance downtime over a defined calendar period.	***Scheduled Maintenance Rate (%) = Scheduled Maintenance Time (SM) / Calendar Time (CT)***
3	**Unscheduled Maintenance Rate**	%	The percentage of Unscheduled Maintenance downtime over a defined calendar period.	***Unscheduled Maintenance Rate (%) = Unscheduled Maintenance Time (UM) / Calendar Time (CT)***
3	**Non-Maintainable Downtime Rate**	%	The percentage of Non-Maintainable downtime over a defined calendar period.	***Non-Maintainable Downtime Rate (%) = Non-Maintainable Downtime (NMD) / Calendar Time (CT)***
3	**Scheduled Maintenance Efficiency**	%	The percentage of Scheduled Maintenance downtime out of the total asset downtime.	***Scheduled Maintenance Efficiency (%) = Scheduled Maintenance Time (SM) / Downtime (DT)***
3	**Unscheduled Maintenance Ratio**	%	The percentage of Unscheduled Maintenance downtime out of the total asset downtime.	***Unscheduled Maintenance Ratio (%) = Unscheduled Maintenance Time (UM) / Downtime (DT)***
3	**Non-Maintainable Downtime Ratio**	%	The percentage of Non-Maintainable downtime out of the total asset downtime.	***Non-Maintainable Downtime Ratio (%) = Non-Maintainable Downtime (NMD) / Downtime (DT)***

6 DOWNTIME & DELAY EVENT CLASSIFICATION PROCESS

Table 3 below outlines the process for classifying downtime and delay events according to the TUM definitions. A cause & effect (C&E) matrix may be developed in conjunction with the implementation of a time & delay accounting system in order to support this process.

Table 3. Downtime & Delay Classification Process

TIME CATEGORY	DEFINITION	PROCESS
Scheduled Maintenance (SM)	• The equipment / asset is down for planned activities that were scheduled and approved in the week prior to the start of the current week. • Classification does not change from scheduled maintenance until the asset is available to operate and the maintenance downtime event is fully completed. • Any emerging corrective activities deemed essential to be completed must be quantified and approved as necessary to prevent a breakdown failure if performed outside the current planning week. • These events, in addition to any maintenance that overruns the planned downtime period will be captured in the weekly planning compliance report.	1. Do not move to another category until the asset is available, and scheduled maintenance has been completed 2. On release from scheduled activities, the equipment will either: a. Go directly into an operating time (OT) element, b. Be placed in an operating delay (OD) element, or, c. Be placed in a standby (SB) element.
Unscheduled Maintenance (UM)	• The equipment / asset is out of service due to a breakdown event. • The event was triggered by a fault or condition resulting in an immediate stoppage where the equipment cannot perform the function for which it was designed. • Classification does not change until the asset is available to operate.	1. Do not move to another category until the asset is available. 2. On release from unscheduled activities, the asset will either, a. Go directly into an operating time (OT) element, b. Be placed in an operating delay (OD) element, or, c. Be placed in a standby (SB) element.

TIME CATEGORY	DEFINITION	PROCESS
Non-Maintainable Downtime (NMD)	• The equipment / asset is out of service due to a non-maintainable event. • Circumstances where a maintenance strategy or activity cannot prevent the delay from occurring. • Examples are accident or cyclone damage, or tyre puncture	1. Do not move to another category until the asset is available. 2. On release from non-maintainable activities, the asset will either, a. Go directly into an operating time (OT) element, b. Be placed in an operating delay (OD) element, or, c. Be placed in a standby (SB) element.
Operating Standby (OSB)	• The equipment / asset is available but not operating due to management decision or reasons within management control.	1. Asset remains in the operating standby element until, a. Decision or reason is no longer applicable, and, b. The asset goes directly into an operating time (OT) element or, c. Is reclassified under another time element as required.
External Standby (ESB)	• The equipment / asset is available but cannot be operated for reasons that are out of the immediate influence of operating management control.	1. Asset remains in the external standby element until, a. The reason is no longer applicable and, b. The asset goes directly into an operating time (OT) element, or: c. Is reclassified under another time element as required.
Operating Delay (OD)	• The equipment / asset is utilised but temporarily stopped OR, • Prevented from performing work due to delays that are inherent to the operation or the immediate physical and environmental conditions.	1. Asset remains in the operating delay element until, a. The reason is no longer applicable, and, b. The asset goes directly into a working time element, or, c. Is reclassified under another time element as required
Non-Productive Time (NPT)	• The equipment / asset is utilised but temporarily stopped due to unavoidable activities that do not directly contribute to production but are required to enable continued safe and efficient operation. • E.g. refueling, training, re-handling	1. Asset remains in the non-productive time element until, a. The reason is no longer applicable, and the asset goes directly into a productive time element, or, b. Is reclassified under another time element as required

7 DOWNTIME EVENT CAPTURE

In order to support effective performance management and production optimisation, downtime events are to be captured and categorised to reflect the relevant ***equipment location***, ***cause***, ***effect*** and ***TUM classification***.

The Site Manager (or delegate) is accountable for the implementation of a site-specific cause & effect matrix which complies with the CSI TUM specification and supports the accurate capture of events.

7.1 Event Capture Methods

Time shall be recorded when an asset is considered handed over to the relevant operations and ready to begin its useful life. This time should be recorded via the CSI corporate delay accounting system (e.g. AMPLA), however where it is not possible or deemed not required, the relevant operational manager shall ensure that alternative capture method is developed and documented in their relevant site specific delay accounting process documentation.

Time is to be recorded to at least one (1) minute granularity for all assets, however where not possible, additional considerations shall be made and documented in the relevant site-specific delay accounting process documentation.

7.2 Data Capture Requirements

Time classification is measured at the appropriate level of the Operating Model (see Master Data Management). All time shall be captured such that it equals calendar time. Table 4 below outlines the data required to be captured for each event in order to produce the minimum requirement of time usage model reporting.

Table 4. Downtime Event Data Capture Requirements

Data Required	Datapoint Definition
Production Unit	The highest level of the equipment asset hierarchy where the event occurred (e.g. Crushing Train 1)
Plant Component	The component of the production unit where the event occurred (e.g. Conveyor)
Equipment ID	The ID number of the component (e.g. 3210CV101)
Equipment Type	The specific piece of equipment where the event occurred (e.g. Belt)
Event Start Time	The start time of the event, to the nearest second
Event End Time	The start time of the event, to the nearest second.
Cause	The cause of the event, as defined by the relevant site cause and effect matrix or delay accounting procedures (e.g. 'Awaiting Feed')
Effect	The effect of the event as defined by the relevant site cause and effect matrix or delay accounting procedures (e.g. 'No Production' or 'Reduced Rate')
Classification	The Time Usage Classification as per the CSI TUM specification (outlined in this document)
Comment	Where required, an explanation of the event in text

7.3 Performance Loss

Performance Loss refers to time the equipment is being utilised but is operating below its required rate and/or recovery. It is comprised of 'rate loss' and 'recovery loss':

- **Rate Loss:** is the number of hours the plant fails to operate at its required rate, calculated net of delays.
- **Recovery Loss:** is the number of hours the plant fails to operate at its required recovery. Note that for plants which do not have a reject or tailings stream, such as crushing plants, recovery is always 100% and therefore Recovery Loss is zero.
- Where relevant this type of loss should be captured as the '*effect*' of a downtime or delay event. The relevant Operational Manager shall ensure that their respective performance loss is factored into their delay accounting methods relevant to that specific site.

7.4 Data Integrity

It is the relevant Operational Manager's responsibility to ensure data capture and integrity maintained at all times.

The importance of maintaining data integrity shall be continually reinforced with personnel involved in the gathering, entry, and interpretation of data concerned with CSI Operating performance measurement. Requirements for manual data entry and capture shall be governed by clear procedures and accountability with regular validation checks carried out to ensure compliance. Thereafter, automated data capture systems shall be subject to regular system integrity checks.

The relevant Operational Manager shall ensure these processes and procedures are clear and documented within their own site-specific delay accounting/cause and reason codes procedures and specifications.

8 TIME USAGE MODEL CHANGE PROCESS

8.1 Managing Change

Revisions of this document may be required from time to time in order to align with changes to the project infrastructure, processes or monitoring requirements. The MinRes Change Management Procedure (MRL-QA-PRO-0002) should be followed for any change to this specification in order to ensure all dependencies and risks are mitigated and the change is implemented in a systematic way.

8.2 Time Usage Model Change Process

Changes to an organisation-wide TUM can have significant effects on the ability to report consistently and effectively across an organization, as well as long term detrimental effects to data quality and integrity. For this reason, the sole owner, The Chief Operating Officer (COO) of CSI Mining Services shall hold the final decision as to the level of change management required to be completed by the change requestor in the event a change is requested.

9 DEFINITIONS AND ABBREVIATIONS

9.1 Definitions

Activity	Actions, tasks, or components of a process.
Availability	The ability of a piece of equipment to perform its required function over a given period of time.
Benchmarking	A structured comparison of performance metrics.
Delay	Any kind of interruption in a productive operation.

Event	A definable or describable change in status.
Maintenance	The actions that are intended to preserve an item in, or restore it to, a state in which it can perform its intended function.
May	See **Should**.
Metrics	Measures of process performance.
Must	See **Shall**.
Process	A series of activities that take place in a definite manner in order to convert inputs into outputs.
Service Meter Unit	A measure of the time during which a component is in operation
Shall	Indicates that compliance with a statement is mandatory for compliance with the objective & there is to be no deviation.
Should	Indicates that a statement is recommended to comply with industry best practice but may be deviated from if required.
Time Categories	Subdivisions of time defined by classifying activities, statuses, and events so as to describe their nature in terms of the value and impact on the operation.
Time Elements	Types of activities, delays, equipment statuses, and events, which are classified into time categories.
Utilisation	A measure of the time a particular piece of equipment is being used.

9.2 Abbreviations

AA	AMG Availability
AMG	Asset Management Group
AS	Australian Standards
AT	Available Time
AU	Asset Utilisation
CT	Calendar Time
DP	Dynamic Productivity
DT	Downtime
EU	Effective Utilisation
GET	Ground Engaging Tools
GMG	Global Mining Guidelines Group
KPI	Key Performance Indicator

MinRes	*See **MinRes***
MRL	Mineral Resources Limited
NP	Non-Productive Time
OD	Operating Delay
OE	Operating Efficiency
OEM	Original Equipment Manufacturer
OP	Operating Productivity
OT	Operational Technology
OT	Operating Time
OU	Operating Utilisation
PA	Physical Availability
PD	Planned Downtime
PE	Production Effectiveness
PM	Preventative Maintenance
PT	Productive Time
SB	Standby
SBE	External Standby
SBO	Operating Standby
SME	Surface Mining Equipment
SMU	Service Meter Unit
ST	Scheduled Time
UA	Use of Availability
UD	Unplanned Downtime
UT	Unscheduled Time
WT	Working Time